

2020 AMC 10A Problem 14 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 14. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 14

Problem:

Real numbers x and y satisfy $x + y = 4$ and $x \cdot y = -2$. What is the value of

$$x + \frac{x^3}{y^2} + \frac{y^3}{x^2} + y?$$

Answer Choices:

- A. 360
- B. 400
- C. 420
- D. 440
- E. 480

Solution:

$$x + \frac{x^3}{y^2} + \frac{y^3}{x^2} + y = x + \frac{x^3}{y^2} + y + \frac{y^3}{x^2} = \frac{x^3}{x^2} + \frac{y^3}{x^2} + \frac{y^3}{y^2} + \frac{x^3}{y^2}$$

Continuing to combine

$$\frac{x^3 + y^3}{x^2} + \frac{x^3 + y^3}{y^2} = \frac{(x^2 + y^2)(x^3 + y^3)}{x^2 y^2} = \frac{(x^2 + y^2)(x + y)(x^2 - xy + y^2)}{x^2 y^2}$$

From the givens, it can be concluded that $x^2y^2 = 4$. Also,

$$(x + y)^2 = x^2 + 2xy + y^2 = 16$$

This means that

$$x^2 + y^2 = 20$$

Substituting this information into

$$\frac{(x^2 + y^2)(x + y)(x^2 - xy + y^2)}{x^2y^2}$$

we have

$$\frac{(20)(4)(22)}{4} = 20 \cdot 22 = \boxed{(D) 440}.$$

OR

As above, we need to calculate

$$\frac{(x^2 + y^2)(x^3 + y^3)}{x^2y^2}$$

Note that x, y , are the roots of $x^2 - 4x - 2$ and so $x^3 = 4x^2 + 2x$ and $y^3 = 4y^2 + 2y$. Thus

$$x^3 + y^3 = 4(x^2 + y^2) + 2(x + y) = 4(20) + 2(4) = 88$$

where $x^2 + y^2 = 20$ and $x^2y^2 = 4$ as in the previous solution. Thus the answer is

$$\frac{(20)(88)}{4} = \boxed{(D) 440}.$$

Note : $(x^2 + y^2 = (x + y)^2 - 2xy = 20$, and $x^2y^2 = (xy)^2 = 4$)

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